Title

Punch incision versus elliptical excision for epidermal inclusion cysts: Systematic review and meta-analysis (protocol)

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Short running title: Punch vs. excision for EIC

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1. Introduction

Epidermal inclusion cysts (EICs) are common tumors in dermatology practice(Vanhoenacker et al. 2020). In one university hospital setting, approximately 1,000 of the patients with epidermal cyst were diagnosed as epidermal cyst: the tumors occupied approximately 50% of mucocutaneous cysts during 12-months(Kamyab et al. 2020). The tumors are typically well-encapsulated, subepidermal, mobile nodules. Patients who have epidermal inclusion cysts often need the removal of the cysts. The reasons for removal can be include increasing size, bad appearance, foul-smelling discharge, and pain. Elliptical excision and manual, blunt dissection of the cyst wall is the most common procedures for cleaning up such lesions(Suliman 2005). Complete excision has been difficult, however, by a longer scar than certain less invasive techniques(Lee et al. 2006).

Punch incision is an alternative procedure to elliptical excision for removal of epidermal inclusion cyst. The punch is assumed to be an alternative approach to the elliptical excision, but the efficacy of the two procedures has never been systematically reviewed and analyzed. To determine if punch incision is less invasive and with lower recurrence for epidermal inclusion cysts, we will conduct a systematic review and meta-analysis of RCTs assessing punch incision versus elliptical excision on epidermal inclusion cysts.

2. Research question

P: patients who are equal or older than 10 with noninfected epidermal inclusion cysts

I: punch incision

C: elliptical excision

O: recurrence rate at 12 months follow-up, the average operative time (mean), the average length of the postoperative wound (mean), all adverse events

3. Method

We followed the Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 for preparing this protocol(Peters et al. 2015). We will publish this protocol in protocols.io (https://www.protocols.io/).

3.1 Inclusion criteria of the articles for the review

3.1.1 Type of studies

We will include individual and cluster randomized controlled trials (RCTs) that assess punch incision versus elliptical excision on epidermal inclusion cysts. We will not apply language or country restrictions. We will include all literature including published, unpublished articles, abstract of conference, and letters.

We will exclude non-RCTs. We will not exclude studies based on the observation period.

3.1.2 Study participants

Patients who have epidermal inclusion cysts

Inclusion criteria:

Noninfected epidermal inclusion cysts which were clinically diagnosed by physicians

Exclusion criteria:

Infected epidermal inclusion cysts

3.1.3 Intervention

A punch incision procedure with dermal punch biopsy trephines

3.1.4 Control

An elliptical excision

3.2 Type of outcomes

3.2.1 Primary outcomes

1. Recurrence rate

Definition: reported recurrence by participants by visit or telephone interview

Period: during the 12 months follow up period

2. Average operative time (mean)

Definition: procedural time recorded from the time of the first incision to the time of suture closure

Period: at the date of the procedure

3. Average length of the postoperative wound (mean)

Definition: length measured at closure

Period: at the date of the procedure

3.2.2 Secondary outcomes

1. All adverse events

Definition: definition of adverse events was set by the original authors

Incidence proportion of all adverse events

Period: during the follow-up period

3.3 Search method

3.3.1 Electronic search

We will search the following databases:

1. the Cochrane Central Register of Controlled Trials (CENTRAL)
2. MEDLINE via Pubmed
3. EMBASE via PROQUEST

See Appendix 1, 2, and 3 for the search strategies.

3.3.2 Other resources

We will also search the following databases for ongoing or recently finished trials:

1. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP)
2. ClinicalTrials.gov.

See Appendix 4, 5 for the search strategies.

We will sort the reference lists of studies, including international guideline(Stevens et al. 2014) as well as the reference lists of eligible studies and articles citing eligible studies. We will inquire the authors of original studies of unpublished or additional data.

3.4 Data extraction

3.4.1 Selection of the studies

Two independent reviewers (KM and JW) will screen every title and abstract of the articles. Articles extracted by the reviewers will be added to the full-text review. Each reviewer will independently perform study selection. We will make contact with the original authors if there are any disagreements regarding the articles. The two reviewers will compare their lists. If there are any differences in opinion, the differences will be resolved by discussion.

3.4.2 Data extraction and management

Two reviewers (KM and JW) will perform data extraction of the included trials independently using the data collection form, which is pre-checked by two randomly selected studies. The collected data will contain information on study design, study population, interventions and outcomes, and results. Any disagreements through the data extraction will be resolved by discussion.

3.5 Risk of bias

Two reviewers (KM and JW) will assess the risk of bias separately. We will use Version 2 of the Cochrane risk-of-bias tool for randomized trials (RoB 2) tool(Higgins JPT 2020). Any disagreements through the risk of bias assessment will be resolved by discussion.

3.6 Assessment of the treatment effect

We will perform a meta-analysis and calculated the relative risk (RR) with 95% confidence intervals (CI) for the following binary outcome: Recurrence rate.

We will perform a meta-analysis and calculate the mean difference (MD) and 95% Cl about the following continuous variables: mean operative time, mean length of the postoperative wound.

We will summarize adverse events based on the definition by the original articles, but we will not perform the meta-analysis.

3.7Handling of missing data

3.8 Unit of analysis issues

Clustering at the level of the enrolled units in cluster-randomized studies.

In dealing with cluster-RCTs, for dichotomous data, we will apply the design effect and calculate effective sample size and number of events using the intra-cluster correlation coefficient (ICC) among each unit and the average cluster size, as described in the Cochrane Handbook(Higgins JPT 2020). If the ICC has not been reported, we will use the ICC of a similar study as a substitute. For continuous data, only the sample size will be reduced; means and standard deviation will remain unchanged(Higgins JPT 2020).

3.8.1 Missing participants

For dichotomous data

We will perform the intention-to-treat (ITT) analysis for all dichotomous data as much as possible. We will also include missing participants for analysis.

For continuous data

We will not impute missing data based on the recommendation by the Cochrane handbook(Higgins JPT 2020). We will perform a meta-analysis of the available data in the original study.

3.8.2Missing values

We will ask for missing values from the original authors.

3.8.3Missing statistics

In case the original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman(Suliman 2005). If we don't know these values when we contact the authors, the standard deviation will be calculated by confidence interval and t-value based on the method by Cochrane handbook(Higgins JPT 2020), or validated method (Furukawa et al. 2006). We will analyze the validity of these methods by using sensitivity analysis.

3.9 Assessment of heterogeneity

We will evaluate the statistical heterogeneity by visual inspection of the forest plots and calculating the I^2 statistic (I^2 values of 0% to 40%: might not be important; 30%to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity)(Higgins JPT 2020). When there is substantial heterogeneity (I2> 50%), we will consider the reason for the heterogeneity about the primary outcomes. Cochrane Chi2test (Q-test) will be performed for I^2 statistic, and P value less than 0.10 will be defined as statistically significant.

3.10 Assessment of reporting bias

We will search for the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform an expansive literature search for unpublished trials. We will assess the potential publication bias by visual inspection of the funnel plot. Egger test will be also performed. We will not conduct the Funnel plot and Egger test in case that we find less than 10 trials or trials which have similar sample sizes(Higgins JPT 2020).

3.11 Meta-analysis

A meta-analysis will be performed using Review Manager software (RevMan 5.4.1.). We will use a random-effects model.

3.12 Subgroup analysis

To clarify the influence of effect modifiers on results, we will conduct the subgroup analyses of the primary outcomes on the following factors when sufficient data are available.

1. (For participants) Age: subgroups will be separated younger than 65 years and older than or equal 65 years

2. (For participants) Diabetes mellitus (DM): subgroups will be having DM and not-having DM.

3. (For participants) Original size: subgroups will be larger than or equal to 10mm and smaller than 10mm.

4. (For intervention) Type of surgeon: subgroups will be certified surgeon and trainee.

3.13 Sensitivity analysis

We will perform sensitivity analysis for the primary outcome to assess whether the results of the review are robust to the decisions made during the review process. We plan to examine the effects of the review findings of:

1. Exclusion of studies using imputed statistics.
2. Only the participants who complete the study with complete data

4. Summary of findings table

Summary of findings table will be made for the following outcome based on the Cochrane Handbook(Higgins JPT 2020).

We will include grading to evaluate the quality of evidence-based on the GRADE (Grading of Recommendations Assessment, Development, and Evaluation) approach for each Summary of findings table(Higgins JPT 2020).

1. Recurrence rate
2. Mean operative time
3. Mean length of the postoperative wound
4. All adverse events

5. Conflict of Interest

The authors declare no conflicts of interest.

Appendix 1: CENTRAL search strategy

([mh "Epidermal Cyst"] OR ((Epidermal:ti,ab OR Sebaceous:ti,ab OR Epidermoid:ti,ab OR Pilar:ti,ab) AND cyst*:ti,ab)) AND punch:ti,ab

Appendix 2: MEDLINE (via PubMed)search strategy

#1 "Epidermal Cyst"[Mesh]

#2 Epidermal[tiab]

#3 Sebaceous[tiab]

#4 Epidermoid[tiab]

#5 Pilar[tiab]

#6 Cyst*[tiab]

#7 (#2 OR #3 OR #4 OR #5) AND #6

#8 #1 OR #7

#9 punch[tiab]

#10 #8 AND #9

Appendix 3: EMBASE search strategy

S1 (EMB.EXACT.EXPLODE("epidermoid cyst"))

S2 ab(Epidermal OR Sebaceous OR Epidermoid OR Pilar) OR ti(Epidermal OR Sebaceous OR Epidermoid OR Pilar)

S3 ab(cyst OR cysts) OR ti(cyst OR cysts)

S4 S3 AND S2

S5 S4 OR S1

S6 (ab(punch) OR ti(punch))

S7 S6 AND S5

Appendix 4: ICTRP search strategy

(((Epidermal OR Sebaceous OR Epidermoid OR Pilar) AND (cyst OR cysts)) AND punch)

Appendix 5: ClinicalTrials.gov search strategy

Status: Recruiting, Not yet recruiting, Available, Active, not recruiting, Completed, Enrolling by invitation, Suspended, Terminated, Withdrawn, No longer available, Temporarily not available, Approved for marketing, Unknown status

Study type: Interventional Studies

Condition or disease: ((Epidermal OR Sebaceous OR Epidermoid OR Pilar) AND (cyst OR cysts))

Intervention: punch

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